Electrochimica Acta 111 (2013) 144–151

Contents lists available at ScienceDirect

Electrochimica Acta

journal homepage: www.elsevier.com/locate/electacta

The mixed former effect in lithium borophosphate oxynitride thin film electrolytes for all-solid-state micro-batteries

Yongsub Yoon, Chanhwi Park, Junghoon Kim, Dongwook Shin *

Division of Materials Science & Engineering, Hanyang University, 17 Haengdang-dong, Seongdong-gu, Seoul 133-791, Republic of Korea

ARTICLE INFO

Article history:
Received 8 February 2013
Received in revised form 2 August 2013
Accepted 9 August 2013
Available online 20 August 2013

Keywords:
Mixed former effects
Lithium borophosphate oxynitride
Li-ion conducting electrolytes
All-solid-state thin film batteries

ABSTRACT

The mixed former effect of boron and phosphorus is applied to improve chemical stability and electrochemical property of thin film electrolyte based on lithium phosphorus oxynitride (LiPON). This thin film with new composition 'lithium borophosphate oxynitride (LiBPON)' is deposited by RF magnetron sputtering with double targets of Li_3PO_4 and Li_3BO_4. When boron content is varied by controlling the RF power on Li_3PO_4 and Li_3BO_4 targets, boron ions successfully replace the four-coordinated phosphorus. Raman spectra and N 1s XPS spectra of this new electrolyte reveal that three-coordinated nitrogen atoms (P—N < P) are dominant in glass network, which distinguishes from LiPON films without the B—P mixed former effect. The improved electrochemical properties and chemical stability of LiBPON film endow all-solid-state thin film cells ($LiCoO_2$/LiBPON/Li) excellent cycling performances with the initial capacity of 34.5 μAh cm^{-2} and the cycling efficiency and 95.3% of its initial capacity.

1. Introduction

In order to realize all-solid-state thin film batteries, numerous candidates for the solid electrolyte material have been investigated. Among them, lithium phosphorus oxynitride (LiPON) thin film glass has been vigorously investigated by the group in Oak Ridge National Lab [1–3]. LiPON prepared by RF-sputtering from Li_3PO_4 target under N_2 atmosphere is known as the best material due to its chemical stability compared to other lithium oxide or sulfide based glasses although its moderate ionic conductivity of $1–3 \times 10^{-6}$ S cm^{-1} at room temperature and activation energy of 0.55 eV are not satisfactory [2,4,5] for all-solid-state battery. A major improvement in chemical durability and electrochemical properties has been attributed to the increased cross-linking between the chains of PO_4 tetrahedron due to the substitution of non-bridging oxygen ions in glass network by doubly (P—N=P) and triply (P—N < P) coordinated nitrogen [6]. Also, based on the Anderson–Stuart model, it was reported that the replacement by P—N bond in the oxynitride glass makes the ionic conductivity and the activation energy improved by decreasing the electrostatic energy of P—O bonding [7]. In summary, there is a solid evidence proving that P—N < P cross-linked structure unit is more responsible for the increase in lithium mobility than P—N=P structural unit. This experimental observation opens the possibility to improve the ionic conductivity if one finds the method to increase the concentration of three-coordinated nitrogen bonds in glass network.

Numerous studies are found in literature showing the relationship between this structural change and the electrochemical property of LiPON. The contents and the species of nitrogen bonds in glass structure are controlled by sputtering parameters such as RF power [10–12], chamber pressure [13,14], gas flow rate [15,16], substrate temperature [17], and lithium content [18,19]. Even though some discrepancies on the influence of sputter deposition conditions [11,12], most studies have clearly shown that the ionic conductivity increases with the nitrogen content in the glass structure. It has been reported that the ionic conductivities of LiPON thin films grown under various deposition conditions are in the range of $5 \times 10^{-7}–2 \times 10^{-6}$ S cm^{-1} at room temperature.

In this study, we deposited lithium borophosphorus oxynitride (LiBPON) thin films as new solid electrolyte candidate for thin film power devices using RF-magnetron sputtering. Boron was added as a glass network former and the mixed former effect with phosphorous was firstly investigated in the LiPON system. In previous report [20], $Li_2O—B_2O_3—P_2O_5$ glass thin films with mixed former effects showed the improved stability and electrochemical property. This oxide electrolyte has been deposited by controlling the ratio between the network modifier and the glass former using a co-sputtering method, and has exhibited that the maximum conductivity of modified thin film electrolytes was 1.22×10^{-6} S cm^{-1}, close to that of LiPON. From these results, LiBPON structure is expected to show the improved ionic conductivity and electrochemical stability due to the synergetic combination of the B—P mixed former effect and the oxygen–nitrogen substitution. In order

* Corresponding author. Tel.: +82 2 2220 0503; fax: +82 2 2220 4011.
E-mail address: dwshin@hanyang.ac.kr (D. Shin).

0013-4686/$ – see front matter © 2013 Elsevier Ltd. All rights reserved.
http://dx.doi.org/10.1016/j.electacta.2013.08.053

to understand the role of the B—P mixed former on ionic conductivity of the electrolyte thin films, we prepared various LiBPON thin films with different B/P ratio and investigated its correlation with the local structure. Also, all-solid-state thin film batteries using the LiBPON thin films are prepared by RF magnetron sputtering techniques.

2. Experimental

LiBPON thin films were deposited by a RF magnetron sputter with double targets of Li_3PO_4 and Li_3BO_4 (Super Conductor Materials Inc., USA). First, RF power for Li_3PO_4 target was fixed to 50 W, and that of Li_3BO_4 was varied to 0 W, 20 W, and 40 W to investigate B—P mixed former effect. The base pressure of the RF reaction chamber was maintained to 10^{-6} Torr by a cryopump, and the working pressure was controlled to 8 mTorr. The sputtering time of LiPON electrolyte (0 W Li_3BO_4) is fixed with 20 h and that of LiBPON electrolytes is reduced under pure nitrogen atmosphere to make the thickness of deposited electrolyte similar. The nitrogen gas flow rate was fixed to 30 sccm. All thin films were deposited at room temperature without intentional heating of the substrate, and the target-to-substrate distance was 10 cm.

The compositional analysis of Li, B, and P was carried out by an inductively coupled plasma-atomic emission spectroscopy (ICP-AES, Optima-4300 DV, USA) after dissolving the samples in the aqua regia and HCl solution. Also, the B, P, O, and N contents were determined by an Electron Probe Micro Analyzer (EPMA, JEOL JXA-8900R, Japan). The medium range crystal structures of thin films were investigated by a Raman spectroscopy (Dongwoo Optron, Vector-01FX, Korea) and a Fourier transform infrared spectroscopy (FT-IR, IRAffinity-1, Japan). To obtain sufficient Raman signal from glass thin films, a He–Cd laser source with the wavelength of 325 nm was chosen. To measure infrared spectra, the LiBPON thin films were deposited on 13 ϕ KBr pellets. Additional information on the nitrogen composition in the thin films were provided by X-ray photoelectron Spectroscopy (XPS, Escalab 220i-XL, UK) using Al Kα radiation as a excitation source. Because of surface charge-induced peak shifts, C 1s at 285 eV was taken as a reference energy position to correct the shift and the spectra were analyzed using Spectral data processor software (XPS International, Ver. 4.1) in which a nonlinear background is assumed and the fitting peaks of the experimental curve are defined to a combination of Gaussian (80%) and Lorentzian (20%) distributions. The reproducibility was within 5% allowing a relatively accurate comparison of data and trends. In order to avoid any surface contamination of the films, all samples were handled in an argon filled glove box and transferred to other apparatus while sealed in gas-tightened boxes and the oxygen contamination level was very low.

The ionic conductivity of the LiBPON films was determined by AC impedance spectroscopy (Solatron Si1260, USA) with Pt/LiBPON/Pt sandwich structures on the SiO_2/Si substrate with the overlap area of 9 mm^2. The platinum electrodes of ~50 nm thickness were prepared by RF magnetron sputtering under argon atmosphere. The thickness and the deposition rate of the prepared LiBPON thin films were determined by a field emission scanning electron microscope (FE-SEM, Hitachi S4800, Japan) and the cross-sectional micrographs of that are shown in Fig. 1. Dense and crack-free LiBPON thin films with similar thickness were deposited on the SiO_2/Si substrate. All specimens did not any difference in morphology with increasing the RF power of Li_3BO_4 target and showed smooth glassy layer of ~500 nm. The deposited glass thin films were annealed at 100 °C to form good contact with the electrodes. The measurements were made in an inert atmosphere to avoid moisture attack.

The thin film batteries were fabricated onto a 20 mm × 15 mm sized sapphire substrate. After the deposition of a 10-nm-thick tantalum adhesion film, a 100 nm platinum metallic layer was deposited as a current collector and annealed at 700 °C for an hour. $LiCoO_2$ cathode layer of ~1 μm thickness was deposited and sintered at 600 °C for 2 h in oxygen atmosphere to decrease lattice strain and enhance the crystallization of as-deposited films. As a counter current collector, nickel was deposited to thickness of ~100 nm by an E-beam evaporator. Solid electrolyte LiBPON was then prepared over the cathode electrode with thickness of ~2000 nm. Finally, the lithium metal anode layer was deposited on the solid electrolyte and the nickel layer. The prepared thin film full-cells were installed to the cable-equipped bottles in Ar-filled glove box, and were charged and discharged using a cycling tester (Toyo system TOSCAT-3100, Japan). The cells were

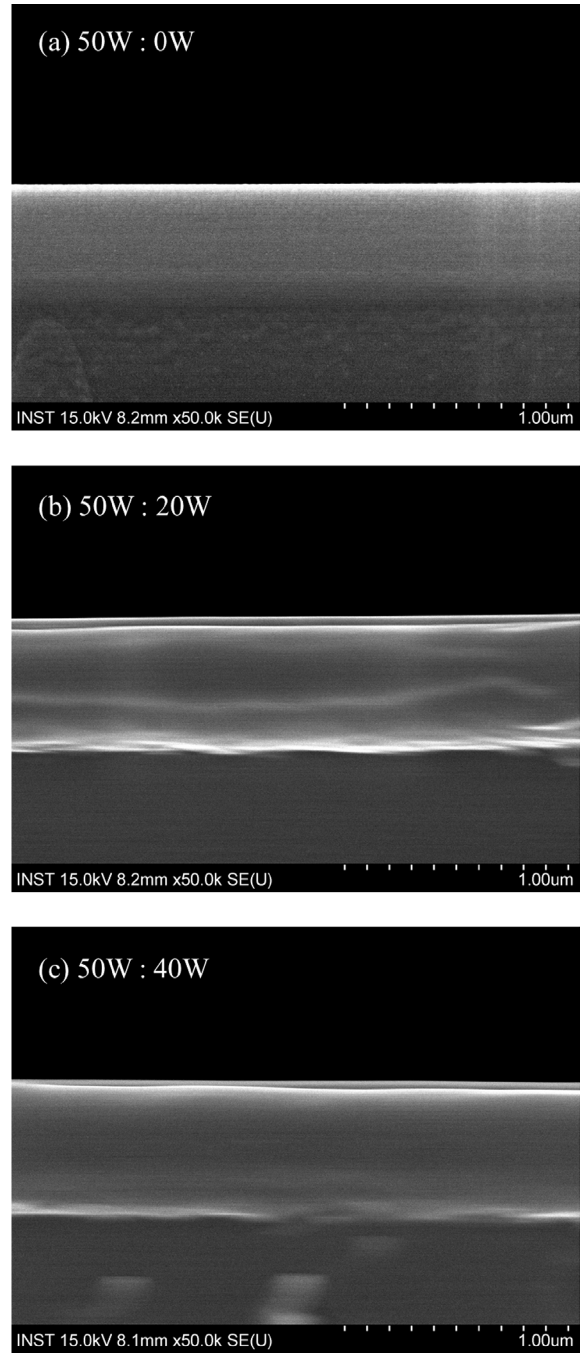

Fig. 1. Cross-sectional FE-SEM images of thin films [(a) 50 W:0 W, (b) 50 W:20 W, (c) 50 W:40 W].

Table 1
Compositional analysis of LiBPON thin films by ICP-AES and EPMA. The composition was controlled by various RF-powers on Li_3PO_4 and Li_3BO_4 targets denoted by 50 W:*x* W [*x* = (a) 0 W, (b) 20 W, and (c) 40 W].

Sample name	RF power (W)		Chamber pressure (mTorr)	Deposit rate (nm/min)	Chemical composition (B + P = 1)	Composition ratio		
	Li_3PO_4	Li_3BO_4				Li/(B + P)	B/(B + P)	N/(O + N)
(a)	50	–	6	1.696	$Li_{2.82}PO_{3.01}N_{0.13}$	2.82	–	0.042
(b)	50	20	8	2.183	$Li_{2.65}B_{0.11}P_{0.89}O_{3.00}N_{0.15}$	2.65	0.11	0.048
(c)	50	40	8	2.615	$Li_{2.07}B_{0.35}P_{0.65}O_{2.45}N_{0.11}$	2.07	0.35	0.041

electrochemically cycled over the voltage range of 3.4 and 4.2 V by the current densities of 10 μA cm^{-2} at room temperature.

3. Results and discussion

To investigate the effect of former mixing in the prepared LiBPON glass electrolytes on the structural modification and the ionic conductivity, the RF power of Li_3PO_4 target was fixed to 50 W, while that of Li_3BO_4 target was varied to (a) 0 W, (b) 20 W, and (c) 40 W, and the sample name was assigned based on the RF power on the Li_3BO_4 target. Table 1 summarizes the deposition conditions and the chemical compositions of the deposited thin films. With increasing the RF power of Li_3BO_4, the Li/B + P ratio decreases from 2.82 to 2.07 while B/B + P ratio increases from 0 to 0.35. This result suggests that the lithium incorporation from Li_3BO_4 target is relatively lower than that from Li_3PO_4. Also, the sample (b) (Li_3BO_4 power; 20 W) shows much higher N content (N/(O + N) = 0.048). Because the ionic conductivity is enhanced by the nitrogen replacement generating non-bridging oxygen by the P—N crosslink, it is expected that sample (b) shows the highest ionic conductivity.

Spectroscopic analysis has been carried out on LiBPON films in order to analyze the local structural variation induced by the nitrogen insertion and its relevance to the electrochemical properties. The measured spectra were compared with those of $Li_2O—B_2O_3$ and $Li_2O—P_2O_5$ amorphous materials. The infrared absorption spectra of LiBPON thin films with various B—P compositions are shown in Fig. 2. Assignments of the IR spectral features were based on the previous reports on phosphate-based glasses [21–23], phosphorus nitride [24], borate-based glasses [25,26], and boron nitride [27,28]. First, in spectrum (a) of the LiPON film prepared without boron, the absorption of the PO_4 and P—O—P groups were observed around 586 cm^{-1}, 940 cm^{-1}, and 1047 cm^{-1}, and the peak of the phosphorus nitride was located at 454 cm^{-1}. As the boron content increases, the boron bands such as BO_4 at 960 cm^{-1} and B—N at 748 cm^{-1}, 1076 cm^{-1}, and 1405 cm^{-1} were increased. An important point to notice here is that the B—O bond peaks of the network structure such as the trigonal BO_3 and the pyroborate B_2O_5 [25], which is dominantly shown from common lithium borate glasses, were not detected. This BO_4^- formation in the $B_2O_3—P_2O_5$ mixed former glass system was similar with results in various reports [29,30]. These experimental results indicate that most of incorporated boron ions replace the phosphorus ions as four-coordinated tetrahedral units and the nitridation was successfully completed in our B—P composition range.

Raman and XPS analysis allow us to look closely into the structural changes, more specifically nitridation mechanism, induced by the mixed former effect. The formation of bonds between phosphorus and nitrogen in the LiPON structure comprises two kinds of bonds, P—N=P and P—N < P, in the midst of the tetrahedral P—O—P chains as explained in the introduction. Fig. 3 displays the Raman spectra obtained from the LiPON thin films (a)–(c) with various B—P compositions. Peak assignment of the spectra is given in Table 2 [31–36]. We observed the P—O bond peaks such as P—O—P chains at 762 cm^{-1}, PO_4^{3-} species at 955 cm^{-1}, and $P_2O_7^{4-}$ species at 1040 cm^{-1} and the intensities of these peaks were slightly decreased with increasing the boron content. At same time,

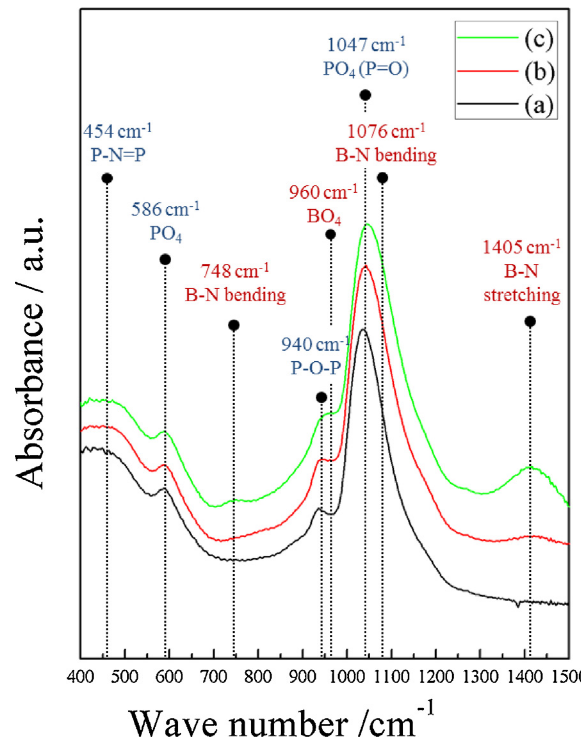

Fig. 2. Infrared absorption spectra of LiBPON thin films with various B—P compositions.

the intensity of the nitridation peak at 620 cm^{-1} and 802 cm^{-1} was dramatically increased from the LiPON thin film (a), and the peak at 620 cm^{-1} assigned to the P—N < P linkages shows the highest intensity at RF power of 50 W:20 W. In case of the P—N=P at 802 cm^{-1}, the relative intensity was difficult to be judged because the peak position overlapped with the B—N peak at 800 cm^{-1}.

Thus, in order to study the structural change in nitrogen linkages in the thin films, the N 1s XPS spectra were measured. The asymmetric N 1s XPS spectra are usually split into two contributions from higher and lower binding energy, which is interpreted as doubly and triply coordinated nitrogen respectively [8–10]. Based on the molecular structure model of Bunker et al. [8], two P—N < P structural units are created in the structure instead of three P—O—P bonds, while two P—N=P structural units are created instead of one P—O—P bond. Compared with two P—N=P structural units, two

Table 2
Peak assignments for Raman spectra of the LiBPON thin films.

Compound	Wave number (cm^{-1})	Assignment	Ref.
$Li_2O—P_2O_5$	762	P—O—P stretching	[31]
	955	P—O bond in PO_4^{3-}	[32,33]
	1040	P—O^- bond in $P_2O_7^{4-}$	[33,34]
LiPON	620	P—N < P bond	[34,35]
	802	P—N=P bond	[34,35]
LiBON	800	B—N bond	[36]

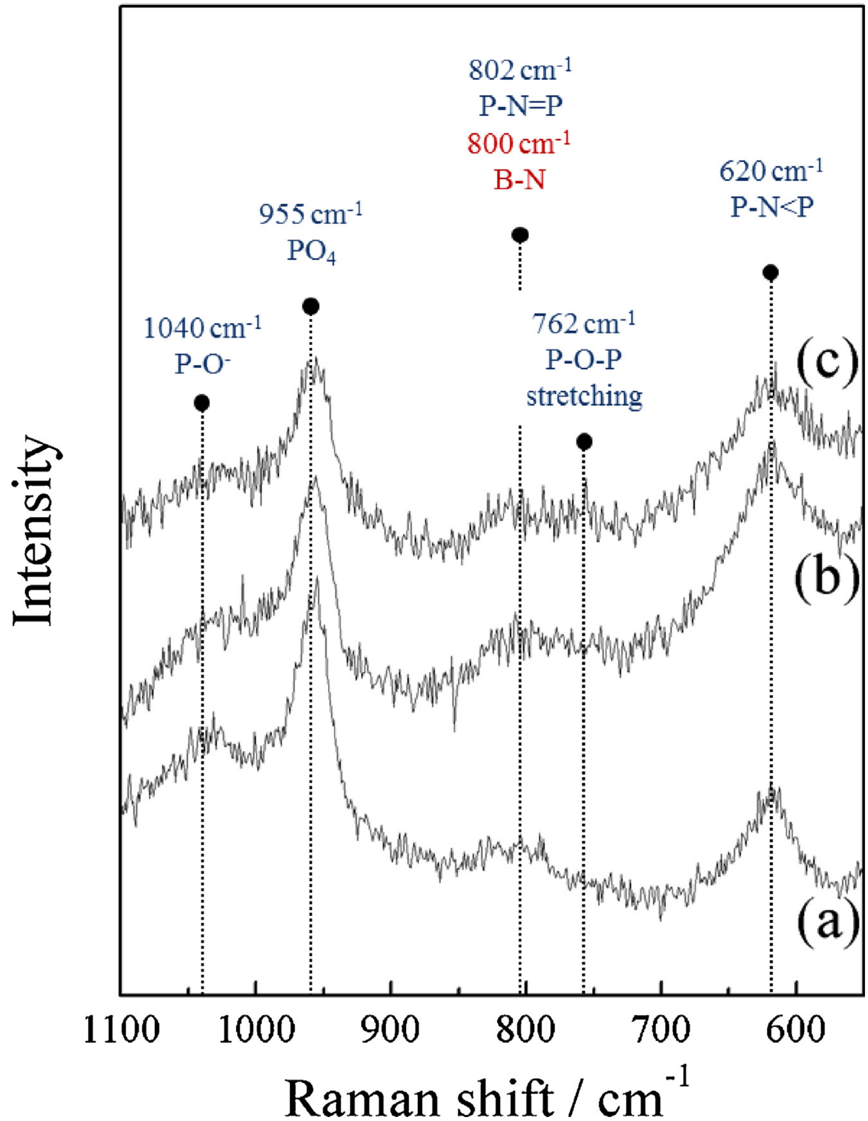

Fig. 3. Raman spectra of LiBPON thin films with various B—P compositions. The RF powers on Li_3PO_4 and Li_3BO_4 targets were denoted by 50 W:x W [x = (a) 0 W, (b) 20 W, and (c) 40 W].

P—N < P structural units remove two more P—O bonds, which are less covalent than P—N bond. Consequently, the decrease of electrostatic energy by P—N < P structural unit is larger than that by P—N═P structural unit.

In Fig. 4, the XPS core level N 1s spectra of LiBPON thin films deposited with various B—P compositions are shown. Asymmetric N 1s spectra of LiBPON films were observed around 398 eV and can be resolved into two peaks. These peaks have been decomposed into two Gaussian–Lorentzian mixed components respectively located at 397.8 eV and 399.4 eV (±0.1 eV) [37]. The low binding energy peak (1) is assigned to two-coordinated nitrogen atoms (P—N═P), while the peak (2) corresponds to three-coordinated nitrogen atoms (P—N < P). In case of the 50 W LiPON specimens, the two-coordinated nitrogen is predominant, while the three-coordinated nitrogen was dominant for the mixed former LiBPON. The quantitative magnitude was calculated from the area of the corresponding photoelectron spectra curve and summarized in Table 3. The three-coordinated nitrogen component shows the highest value of 66% at sputtering condition of 50 W:20 W, and it was also dominant as 58.6% in specimen of 50 W:40 W. The reason for this three-coordinated nitridation in the mixed former LiBPON can be explained by the structural models shown in Fig. 5. This figure compares the nitridations to two- and three-coordinated nitrogen with that of corresponding oxide and the possible variation induced by the mixed former effect. As shown in the left line, LiPON was produced by replacing the P—O—P bridging oxygen bonds to P—N═P or P—N < P and forming the non-bridging oxygen such as P═O or P—O^-···Li^+ from the substitution of oxygen by nitrogen. However, in case of LiBPON, this nitridation mechanism works differently from that in LiPON. In the mechanism of the two-coordinated nitrogen formation, the N═P double bonding removes a non-bridging oxygen P═O in the B—P mixed former glass structure and makes a P—O—P bridging oxygen, which impedes the lithium ion motion by eliminating relatively spacious and loosely bound

Table 3
Calculated ratios of XPS N 1s peaks in the LiBPON thin films deposited at various RF powers (Li_3PO_4:Li_3BO_4).

RF powers		Peak area (%)	
Li_3PO_4	Li_3BO_4	N_{double} (397.8 eV)	N_{triple} (399.4 eV)
50 W	0 W	62.2	37.8
50 W	20 W	34.0	66.0
50 W	40 W	41.4	58.6

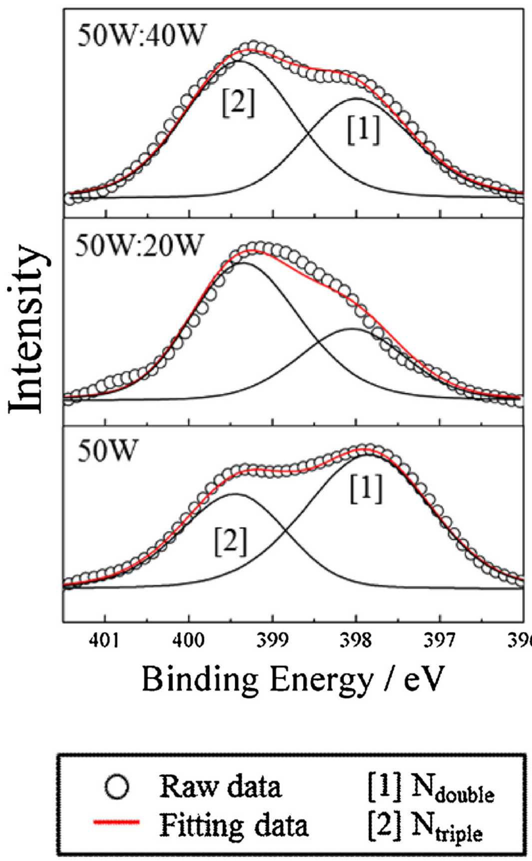

Fig. 4. N 1s XPS of the LiBPON thin films with various B—P compositions. The RF powers of Li_3PO_4 and Li_3BO_4 targets were denoted by 50 W:x W [x = 0 W, 20 W, and 40 W].

Li^+ ion sites. Therefore, the formation of B—N═P in the glass network implicates the reduction of mobile lithium sites. In contrary, the three-coordinated nitrogen (B—N < P) preserves or increases the number of mobile lithium sites such as non-bridging P═O bonding. As XPS results shown in Fig. 4 and Table 3 suggest, this three-coordinated nitrogen (B—N < P) is predominant in LiBPON mixed former electrolyte and this is the reason why we obtained higher ionic conductivity in this glass.

The ionic conductivity of LiBPON thin film electrolytes with various boron contents was measured by an impedance analyzer. The conductivities of various specimens fitted to $T = \sigma_0 \exp(-E_a/RT)$ are shown in Fig. 6(a), and the room temperature conductivities and activation energies are shown in Fig. 6(b) as functions of RF power. As expected from the previously discussed structural variations, the maximum conductivity of 6.88×10^{-7} S cm^{-1} is obtained at the RF powers of 50 W:20 W. However, as the boron content increases by the power of 50 W:40 W, the conductivity was decreased to 2.74×10^{-7} S cm^{-1}. This result may be explained by the fact that the relatively lower lithium content than that of 50 W:20 W makes insufficient non-bridging oxygen sites as shown from the Li/P values in Table 1. Also, the obtained values were not improved in comparison with the previous reports on LiPON structures. In case of LiPON, even though it generally showed the ionic conductivity of about 1–3×10^{-6} S cm^{-1} with the modification of the local structures by controlling sputtering conditions such as RF power, chamber pressure, gas flow rate, and substrate temperature, our LiPON electrolyte deposited by the RF power of 50 W has the conductivity of 2.01×10^{-7} S cm^{-1}. Also, the activation energy was relatively high as 0.653 eV. This low conductivity is due to low nitrogen content in obtained thin films (~30% of the conventional LiPON nitrogen content).

There have been reports on the change in ionic conductivity and low power output from LiPON-based thin film batteries due to the moisture [38–40]. We estimated the chemical durability or the influence of moisture on the ionic conductivity in the LiBPON

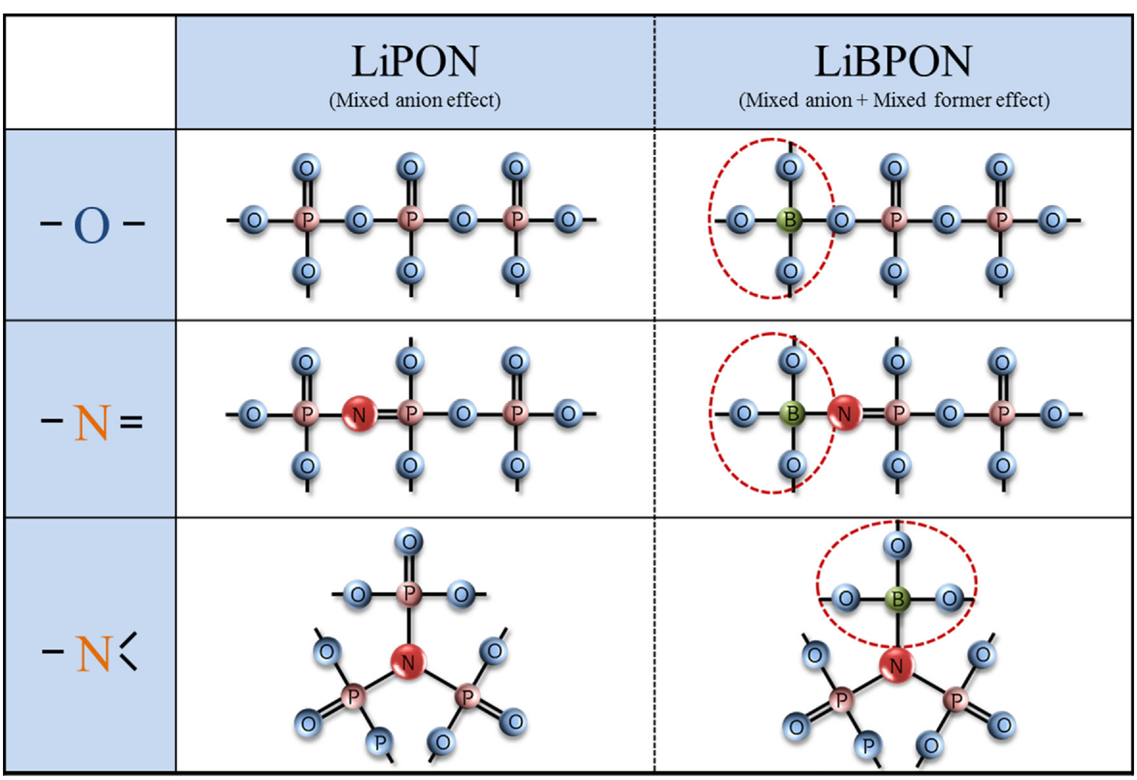

Fig. 5. The comparison of the formation mechanisms of two- and three-coordinated nitrogens in LiPON and LiBPON.

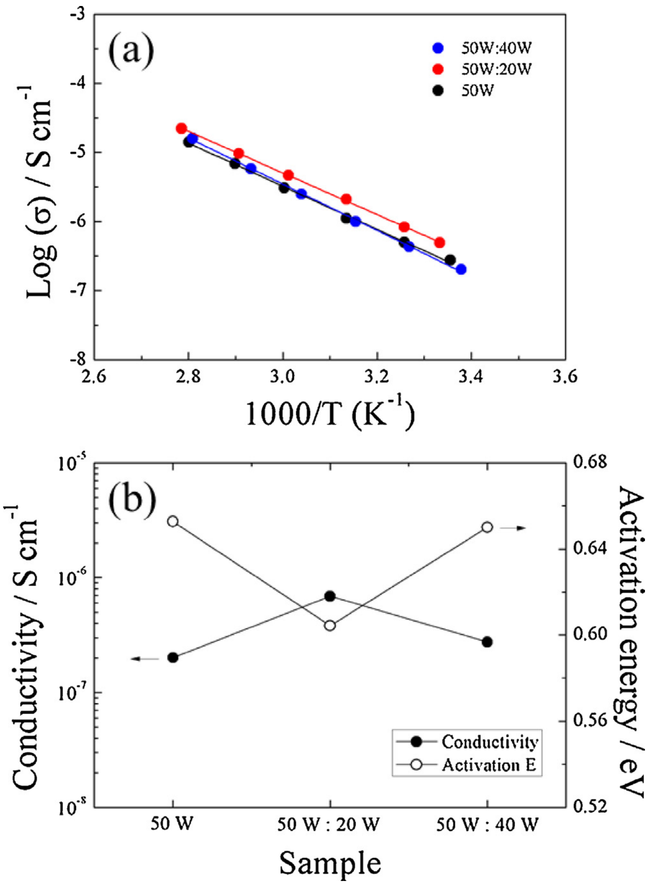

Fig. 6. (a) Arrhenius plots of ionic conductivity and (b) comparison of the conductivity at room temperature and the activation energy of LiBPON films deposited with various RF powers (Li_3PO_4:Li_3BO_4).

thin films. Pt/LiBPON/Pt test samples were kept in ambient air with relative humidity of 50% at room temperature for 40 days, and checked ionic conductivity every 10 days. Fig. 7 shows the variation of Nyquist plots with exposure time. During relatively short storage time of 20 days, all specimen shows typical impedance profile of Li-ion conducting glass electrolytes with Pt electrodes as an ion blocking electrode. The semicircle at high frequency corresponding to resistance of solid electrolyte layer and the linear spike at low frequency corresponding to Li-ion blocking behavior are clearly observed. In case of the LiBPON thin films shown in Fig. 7(b) and (c), impedance plots composed of a semicircle and a

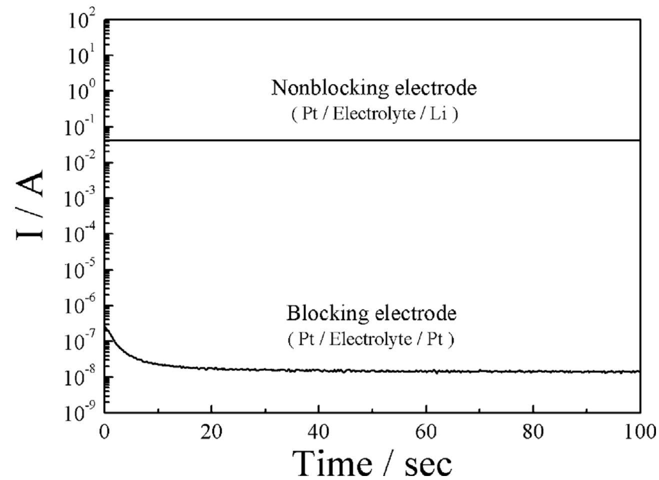

Fig. 8. Time dependent DC current of the LiBPON electrolytes with blocking and non-blocking electrodes. Applied DC potential was 0.1 V.

characteristic capacitive spike maintained its original shape without any severe chemical reactions for 40 days except the increase of electrolyte resistance of 50 W:40 W LiBPON thin films by chemical reaction with the moisture. However, LiPON specimen (a) shows drastic change in the impedance plot. At 30th day, a new resistance component was appeared due to the interfacial reactions of solid electrolyte, and the linear spike at low frequency range originated from the Li-ion blocking behavior of Pt electrode was changed into a semicircle by side reaction between the interfacial layer and Pt electrode. Nimisha et al. [41] also reported that the reaction of electrolyte layer with moisture, oxygen and carbon of air causes the reduction of ionic conductivity and the formation of new interface resistance. This indicates that the reactivity of LiPON specimen is effectively suppressed by the introduction of boron.

The superior stability of 50 W:20 W LiBPON thin films can be explained by the fact that the incorporated boron atoms, which is known as a strong glass network former, make a glass network more rigid. The network connectivity was improved with the formation of bridging oxygen by the B—P mixed former as shown in Fig. 5.

The other important factor for solid electrolyte is electronic insulation or transport number. DC conductivity was measured using lithium metals as non-blocking electrodes and platinum films as blocking electrodes to determine lithium ion transport number of the obtained LiBPON thin film electrolytes. Fig. 8 shows the time dependence of the DC current for the LiBPON thin films

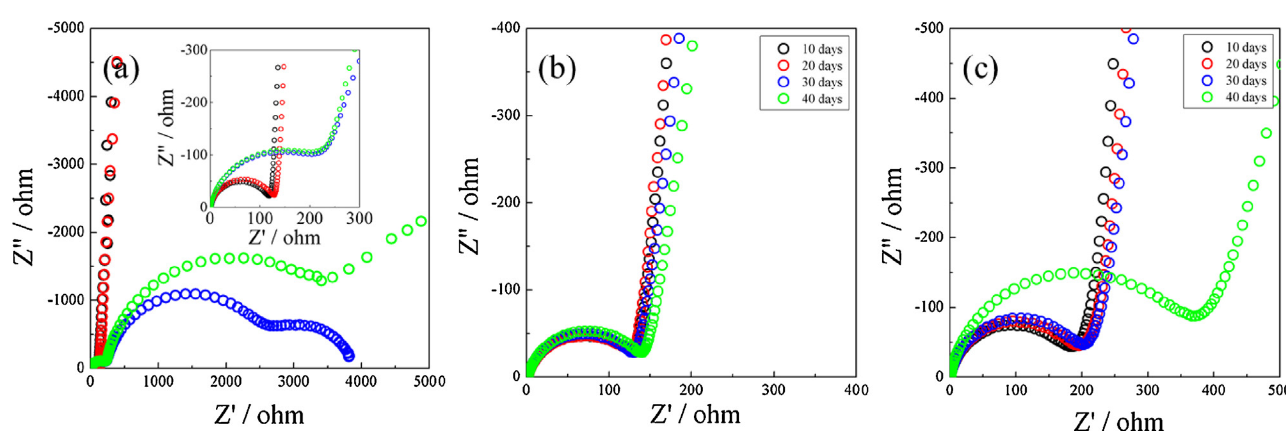

Fig. 7. Nyquist plots obtained from Pt/LiBPON/Pt sandwich thin films when exposed to ambient air with a relative humidity of 50% at a room temperature for 40 h. The LiBPON thin films were prepared by the co-sputtering from Li_3PO_4 target with RF power of 50 W and Li_3BO_4 of (a) 0 W, (b) 20 W, and (c) 40 W.

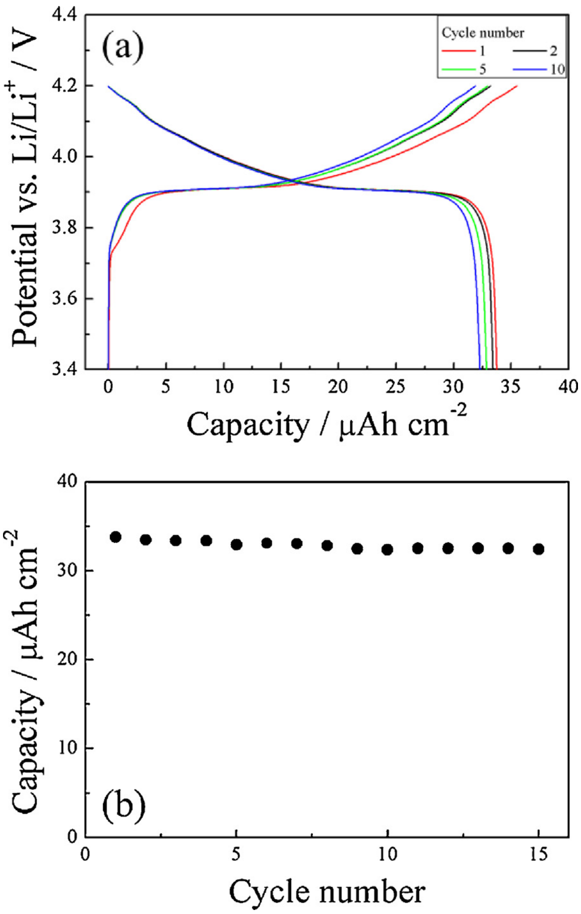

Fig. 9. Cycle behavior of all-solid-state thin film cell Li/LiBPON/$LiCoO_2$ at constant current 10 µA cm^{-2} and cut-off voltage 3.4–4.2 V. (a) Irreversible capacity between 1st and 10th cycles and (b) discharging capacity fading during 15 cycles.

by RF power of 50 W:20 W. A constant DC potential of 0.1 V was applied to the sample. In case of lithium electrodes (non-blocking electrodes), the DC current was almost constant with time, and the conductivity calculated from this steady state value is around 1.628×10^{-6} S cm^{-1}, which almost agrees with the value obtained from the AC impedance measurement using Pt/LiBPON/Pt sandwich electrodes. In case of platinum thin films (the blocking electrodes), a large decay of DC current was initially observed and then reached to a steady state value. The conductivity obtained from the steady state current at 100th sec is 9.326×10^{-12} S cm^{-1}, which are about 6 orders of magnitude lower than that obtained for non-blocking electrode condition. The lithium ion transport number was calculated to 0.9999965 and this suggested that the electronic conduction in the deposited electrolyte is almost negligible.

The LiBPON films were incorporated into a thin film battery consisted of crystallized $LiCoO_2$ cathode and evaporated lithium metal anode. As-deposited battery was heated at 120 °C for improving each interface contact in the cell. Fig. 9(a) shows the charge–discharge curves of a $LiCoO_2$/LiBPON/Li thin film battery with cut-off voltage of 3.4–4.2 V at current density of 10 µA cm^{-2}. This solid electrolyte cell showed lower polarization and higher reversible capacity during the charge–discharge reaction. It shows the clear voltage plateau of ~3.9 V during both charging and discharging. Fig. 9(b) exhibits the cycle-life performance of this cell. At 15th cycles, the cells showed the discharge capacity of 33.2 µAh cm^{-2} and the cycling efficiency was 95.3% of its initial capacity.

4. Conclusions

LiBPON thin films with the B—P mixed former were deposited using a RF magnetron sputter with double targets of Li_3PO_4 and Li_3BO_4. When the boron content is varied by controlling the RF power for Li_3PO_4 and Li_3BO_4, most of added boron ions successfully replaced the phosphorus sites in the B/B + P range of 0–0.35. Raman and XPS spectra for N 1s revealed that three-coordinated nitrogen atoms (P—N < P) were dominant in these new structures, vastly different from the result for the LiPON films without the B—P mixed former effect. As expected from the structural modification, the mixed former LiBPON film deposited with the RF powers of 50 W:20 W showed the maximum conductivity of 6.88×10^{-7} S cm^{-1}. It is required to further increase the nitridation in the films for better conductivity. Also, through the exposure to humid conditions, the improved chemical durability of LiBPON films could be confirmed. All-solid-state thin film cells $LiCoO_2$/LiBPON/Li exhibited excellent cycling performance with the initial capacity of 34.5 µAh cm^{-2} and the cycling efficiency and 95.3% of its initial capacity.

Acknowledgement

This research was supported by a grant from the Fundamental R&D Program for Core Technology of Materials funded by the Korean Ministry of Knowledge Economy (grant no. 10037233).

References

[1] J.B. Bates, N.J. Dudney, G.R. Gruzalski, R.A. Zhur, A. Choudhury, C.F. Luck, J.D. Roberts, Fabrication and characterization of amorphous lithium electrolyte thin films and rechargeable thin-film batteries, J. Power Sources 43 (1993) 103.
[2] X. Yu, J.B. Bates, G.E. Jellison, F.X. Hart, A stable thin-film lithium electrolyte: lithium phosphorus oxynitride, J. Electrochem. Soc. 144 (1997) 524.
[3] J.B. Bates, N.J. Dudney, B. Neudecker, A. Ueda, C.D. Evans, Thin-film lithium and lithium-ion batteries, Solid State Ionics 135 (2000) 33.
[4] C.S. Nimsha, K. Yellareswar Rao, G. Venkatesh, G. Mohan Rao, N. Munichandraiah, Sputter deposited LiPON thin films from powder target as electrolyte for thin film battery applications, Thin Solid Films 519 (2011) 3401.
[5] B. Fleutot, B. Pecquenard, H. Martinez, M. Letellier, A. Levasseur, Investigation of the local structure of LiPON thin films to better understand the role of nitrogen on their performance, Solid State Ionics 186 (2011) 29.
[6] B. Wang, B.S. Kwak, B.C. Sales, J.B. Bates, Ionic conductivities and structure of lithium phosphorus oxynitride glasses, J. Non-Cryst. Solids 183 (1995) 297.
[7] H. Unuma, K. Komiri, S. Sakka, Electrical conductivity and chemical durability in alkali-silicate oxynitride glasses, J. Non-Cryst. Solids 95/96 (1987) 913.
[8] B.C. Bunker, D.R. Tallant, C.A. Balfe, Structure of phosphorus oxynitride glasses, J. Am. Ceram. Soc. 70 (1987) 675.
[9] J.B. Bates, N.J. Dudney, G.R. Gruzalski, R.A. Zuhr, A. Choudry, C.F. Luck, J.D. Robertson, Electrical properties of amorphous lithium electrolyte thin films, Solid State Ionics 53–56 (1992) 647.
[10] Z. Hu, D. Li, K. Xie, Influence of radio frequency power on structure and ionic conductivity of LiPON thin films, Bull. Mater. Sci. 31 (2008) 681.
[11] C.H. Choi, W.I. Cho, B.W. Cho, H.S. Kim, Y.S. Yoon, Y.S. Tak, Radio-frequency magnetron sputtering power effect on the ionic conductivities of LiPON films, Electrochem. Solid-State Lett. 5 (2002) A14.
[12] Y. Hamon, A. Douard, F. Sabary, C. Marcel, P. Vinatier, B. Pecquenard, A. Levasseur, Influence of sputtering conditions on ionic conductivity of LiPON thin films, Solid State Ionics 177 (2006) 257.
[13] H.Y. Park, S.C. Nam, Y.C. Lim, K.G. Choi, K.C. Lee, G.B. Park, S.R. Lee, H.S. Park, S.B. Cho, Effects of sputtering pressure on the characteristics of lithium ion conductive lithium phosphorous oxynitride thin film, J. Electroceram. 17 (2006) 1023.
[14] Z. Hu, K. Xie, D. Wei, N. Ullah, Influence of sputtering pressure on the structure and ionic conductivity of thin film amorphous electrolyte, J. Mater. Sci. 46 (2011) 7588.
[15] B. Fleu tot, B. Pecquenard, H. Martinez, M. Letellier, A. Levasseur, Investigation of the local structure of LiPON thin films to better understand the role of nitrogen on their performance, Solid State Ionics 186 (2011) 29.
[16] Y.G. Kim, H.N.G. Wadley, The influence of the nitrogen-ion flux on structure and ionic conductivity of vapor deposited lithium phosphorus oxynitride films, J. Power Sources 196 (2011) 1371.
[17] S. Jacke, J. Song, L. Dimesso, J. Brotz, D. Becker, W. Jaegermann, Temperature dependent phosphorous oxynitride growth for all-solid-state batteries, J. Power Sources 196 (2011) 6911.

[18] N. Suzuki, S. Shirai, N. Takahashi, T. Inaba, T. Shiga, A lithium phosphorous oxynitride (LiPON) film sputtered from unsintered Li_3PO_4 powder target, Solid State Ionics 191 (2011) 49.
[19] N. Suzuki, T. Inaba, T. Shiga, Electrochemical properties of LiPON films made from a mixed powder target of Li_3PO_4 and Li_2O, Thin Solid Films 520 (2012) 1821.
[20] Y. Yoon, C. Park, J. Kim, D. Shin, Characterization of lithium borophosphate glass thin film electrolytes deposited by RF-magnetron sputtering for micro-batteries, Solid State Ionics 225 (2012) 636.
[21] B.V.R. Chowdari, K.F. Mok, J.M. Xie, R. Gopalakrishnan, Spectroscopic and electrical studies of silver sulfophosphate glasses, J. Non-Cryst. Solids 160 (1993) 73.
[22] N.S. Roh, S.D. Lee, H.S. Kwon, Effects of deposition condition on the ionic conductivity and structure of amorphous lithium phosphorus oxynitrate thin film, Scripta Mater. 42 (2000) 43.
[23] W.S. Chen, E.A. Monroe, R.A. Condrate, Y.M. Guo, An investigation of the hardening process for several phosphate glass-containing cements, J. Mater. Sci. Mater. Med. 4 (1993) 111.
[24] Q. Guo, Q. Yang, L. Zhu, C. Yi, Y. Xie, Large-scale synthesis of amorphous phosphorus nitride imide nanotubes with high luminescent properties, J. Mater. Res. 20 (2005) 325.
[25] E.I. Kamitsos, A.P. Patsis, M.A. Karakassides, G.D. Chryssikos, Infrared reflectance spectra of lithium borate glasses, J. Non-Cryst. Solids 126 (1990) 52.
[26] A.F.L. Almeida, D. Thomazini, I.F. Vasconcelos, M.A. Valente, A.S.B. Sombra, Structural studies of lithium triborate (LBO–LiB_3O_5) in borophosphate glass-ceramics, Int. J. Inorg. Mater. 3 (2001) 829.
[27] A. Phani, Thin films of boron nitride grown by CVD, Bull. Mater. Sci. 17 (1994) 219.
[28] H. Jensen, U.M. Jensenqnxkrdu, G. Sorensen, Deposition of boron-nitride films by nitrogen sputtering from a boron-metal target, Appl. Phys. Lett. 66 (1995) 1489.
[29] K.I. Cho, S.H. Lee, K.H. Cho, D.W. Shin, Y.K. Sun, Li_2O–B_2O_3–P_2O_5 solid electrolyte for thin film batteries, J. Power Sources 163 (2006) 223.
[30] K. Cho, T. Lee, J. Oh, D. Shin, Fabrication of Li_2O–B_2O_3–P_2O_5 solid electrolyte by aerosol flame deposition for thin film batteries, Solid State Ionics 178 (2007) 119.
[31] Y. Kowada, M. Tatsumisago, T. Minami, Raman spectra of rapidly quenched glasses in the systems Li_3BO_3–Li_4SiO_4–Li_3PO_4 and $Li_4B_2O_5$–$Li_6Si_2O_7$–$Li_4P_2O_7$, J. Phys. Chem. 93 (1989) 2147.
[32] K. Chung, W.S. Kim, Y.K. Choi, Lithium phosphorous oxynitride as a passive layer for anodes in lithium secondary batteries, J. Electroanal. Chem. 566 (2004) 263.
[33] X. Yu, J.B. Bates, G.E. Jellison Jr., F.X. Hart, A stable thin-film lithium electrolyte: lithium phosphorus oxynitride, J. Electrochem. Soc. 144 (1997) 524.
[34] T. Pichonat, C. Lethien, N. Tiercelin, S. Godey, E. Pichonat, P. Roussel, M. Colmont, P.A. Rolland, Further studies on the lithium phosphorus oxynitride solid electrolyte, Mater. Chem. Phys. 123 (2010) 231.
[35] L. Popovic, B. Manoun, D. de Waal, M.K. Nieuwoudt, J.D. Comins, Raman spectroscopic study of phase transitions in Li_3PO_4, J. Raman Spectrosc. 34 (2003) 77.
[36] S. Manorama, G.N. Chaudhari, V.J. Rao, Growth of BN by hot filament assisted electron beam deposition, J. Phys. D 26 (1993) 1793.
[37] A. Le Sauze, L. Montagne, G. Palavit, R. Marchand, Nitridation of alkali metaphosphate glasses: a comparative structural analysis of the Na–P–O–N and Li–Na–P–O–N systems, J. Non-Cryst. Solids 293–295 (2001) 81.
[38] K. Nagata, T. Nanno, All solid battery with phosphate compounds made through sintering process, J. Power Sources 174 (2007) 832.
[39] W.C. West, J.F. Whitacre, V. White, B.V. Ratnakumar, Fabrication and testing of all solid-state micro-scale lithium batteries for microspacecraft applications, J. Micromech. Microeng. 12 (2002) 58.
[40] W.Y. Liu, Z.W. Fu, Q.Z. Qin, A sequential thin-film deposition equipment for in-situ fabricating all-solid-state thin film lithium batteries, Thin Solid Films 515 (2007) 4045.
[41] C.S. Nimisha, G.M. Rao, N. Munichandraiah, G. Natarajan, D.C. Cameron, Chemical and microstructural modifications in LiPON thin films exposed to atmospheric humidity, Solid State Ionics 185 (2011) 47.